Abstract

Abstract I

Active Inference offers a unifying account of perception, learning, and action under the free energy principle (Friston 2010), yet the generative models at its core are still communicated ad hoc: scattered across prose descriptions, bespoke notebooks, and framework-specific code that rarely agree. This fragmentation makes published models hard to reproduce, compare, or port between tools, and it raises the barrier for newcomers learning the formalism (Parr et al. 2022). GeneralizedNotationNotation (GNN) addresses this gap with a standardized, human- and machine-readable text language for specifying Active Inference generative models, paired with a 25-step processing pipeline that transforms a single specification into validation, visualization, simulation, and analysis artifacts (Smékal and Friedman 2023). GNN's "Triple Play" treats each model as three coordinated views — a textual specification, graphical visualizations, and executable cognitive models — so that one source yields consistent outputs across modalities. The framework spans 9 model families and 9 registered rendering backends, with explicit gates for cross-format semantic fidelity and cross-framework reliability. GNN 3.2.0 layers safe-by-design long-running orchestration on top of this language and pipeline — durable observation streams, resumable run sessions, and auditable container plans that generate, validate, and replay data only, with no live infrastructure mutation. We describe the language, the pipeline architecture, and the validation that confirms specifications round-trip faithfully across formats and render across multiple simulation backends — with coverage gaps recorded as explicit profiled-unsupported statuses rather than silently omitted — establishing GNN as reproducible, interoperable infrastructure for communicating Active Inference models.